

IDE - Introduction to Single-Variable Calculus I

Final Exam

9:00 am - 11:00 am

* Show explicitly the intermediate steps to show how you reach your solutions.

Prob 1 (10) Suppose g is an even function and define a composite function h by $h = f \circ g$. Show that $h(-x) = h(x)$. This result implies that the composite function h is always an even function.

Prob 2 (10) Let $f(x) = \ln(x + \sqrt{x^2 + 1})$. Find the inverse function of f .

Prob 3 (10) Prove that $\lim_{x \rightarrow 0} \left[x \sin\left(\frac{\pi}{x}\right) \right] = 0$ by using the Squeeze theorem.

Prob 4 (20) Find the limit of

$$\lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{\sin x - \cos x}$$

by using the following two different methods;

- (a) l'Hospital's rule
- (b) Any method other than l'Hospital's rule.

Prob 5 (10) Prove the following identity;

$$\sin^{-1}(\tanh x) = \tan^{-1}(\sinh x)$$

(Hint) Let $X = \sin^{-1}(\tanh x)$. Then, express $\tan^2 X$ in terms of x .

Prob 6 (10) Under the linear approximation in the neighborhood of $x = 0$, show that $f(x)$ is approximated by

$$f(x) = \frac{1}{(1 + 2x)^4} \approx 1 - 8x$$

Prob 7 (10) We want to evaluate the area below $y = x^3$ from $x = 1$ and 2. Find the Riemann sum S_n of the area by dividing the region $[1, 2]$ into n rectangles. Taking the limit of S_n as $n \rightarrow \infty$, find the exact area S .

(Hint) For your reference,

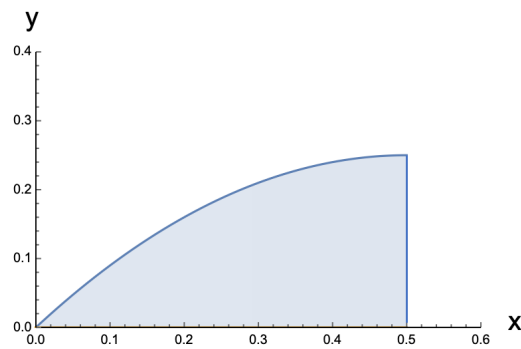
$$\begin{aligned} \sum_{k=1}^n k &= 1 + 2 + \cdots + n = \frac{1}{2}n(n+1) \\ \sum_{k=1}^n k^2 &= 1^2 + 2^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1) \\ \sum_{k=1}^n k^3 &= 1^3 + 2^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 \end{aligned}$$

You may use all formulas above, yet actually you don't need all.

(Continue to the back)

Prob 8 (20) By using the two different methods, cylindrical shells and slices (slabs), calculate the volume V of the solid obtained by rotating the region bounded by $y = x - x^2$ ($0 \leq x \leq 1/2$) and $y = 0$ (the shaded region of the figure below) about the y -axis.

- (a) Find the volume V by using the method of cylindrical shells. Namely, construct first the volume $\Delta V(x)$ of the cylindrical shell at position x with thickness Δx . Then, find the entire volume V by integrating $\Delta V(x)$ over x .
- (b) Find the volume V by slicing the solid of revolution by horizontal plates. Namely, construct first the volume $\Delta V(y)$ of the plate (slab) at position y with thickness Δy . Then, find the entire volume V by integrating $\Delta V(y)$ over y .



(End of the problems)

Single-Variable Calculus I (Final)

Prob #1

$$g \text{ is even} \Rightarrow g(-x) = g(x)$$

$$\text{Then, } h(-x) = f(g(-x)) = f(g(x)) = h(x)$$

Hence, $h(x)$ is an even function $\square$

Prob #2

$$\text{Let } y = \ln(x + \sqrt{x^2 + 1}) \Rightarrow e^y = x + \sqrt{x^2 + 1}$$

$$\begin{matrix} \text{||} \\ \times \end{matrix}$$

$$(x - x)^2 = (\sqrt{x^2 + 1})^2 \Rightarrow x = \frac{x^2 - 1}{2x} = \frac{x - x^{-1}}{2}$$

$$\therefore x = \frac{e^y - e^{-y}}{2} = \sinh y = f^{-1}(y)$$

$$\text{Interchanging } x \leftrightarrow y, \quad f^{-1}(x) = \sinh x \quad \square$$

Prob #3

$$\text{Using } -1 \leq \sin\left(\frac{\pi}{x}\right) \leq 1 \quad (x \neq 0)$$

$$\therefore -x \leq x \sin\left(\frac{\pi}{x}\right) \leq x$$

$f(x) = \pm x$ is continuous around $x = 0$.

$$\therefore \lim_{x \rightarrow 0} (\pm x) = 0,$$

$$\therefore \lim_{x \rightarrow 0} x \sin\left(\frac{\pi}{x}\right) = 0 \quad \square$$

Prob #4

$$(a) \frac{(1 - \tanh x)'}{(\sin x - \cos x)'} = \frac{-\frac{1}{\cosh^2 x}}{\cos x + \sin x} \xrightarrow{x \rightarrow \frac{\pi}{4}} \frac{-2}{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}} = -\sqrt{2} //$$

$$(b) \lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tanh x}{\sin x - \cos x} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{1}{\cos x} \cdot \frac{1 - \tanh x}{\tanh x - 1}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{(-1)}{\cos x} = -\sqrt{2} //$$

Prob #5

$$\text{Let } x = \sinh^{-1}(\tanh x) \Rightarrow \sin x = \tanh x.$$

$$\tanh^2 x = \frac{1}{\cosh^2 x} - 1 = \frac{\sinh^2 x}{1 - \sinh^2 x} = \frac{\tanh^2 x}{1 - \tanh^2 x}$$

$$= \tanh^2 x \cdot \cosh^2 x = \sinh^2 x$$

$$\text{Using that } \tanh x \geq 0, \quad \tanh x = \sinh x$$

$$\therefore x = \tanh^{-1}(\sinh x) \quad \square$$

Prob #6

The linear approx around $x=0$ is given by

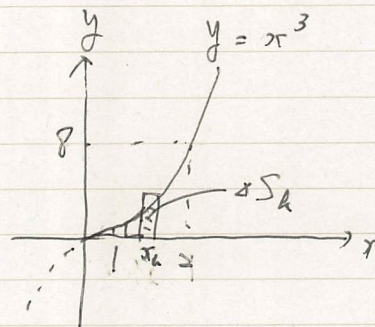
$$f(x) = f(0) + x f'(0)$$

$$f'(x) = -4(1+2x)^{-5} \cdot 2 = -\frac{8}{(1+2x)^5}$$

$$\therefore f(x) = 1 - 8x //$$

Prob #7

$$\begin{aligned}
 \Delta S_k &= f(x_k) \Delta x \\
 &= f\left(1 + \frac{k}{n}\right) \Delta x \quad (\because x_k = 1 + \frac{k}{n}) \\
 &= \left(1 + \frac{k}{n}\right)^3 \cdot \frac{1}{n}
 \end{aligned}$$

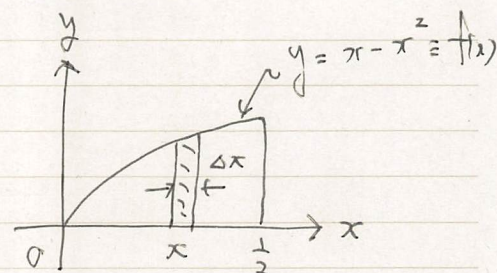


$$\begin{aligned}
 \text{Riemann sum} \quad S_n &= \sum_{k=1}^n \Delta S_k = \sum_{k=1}^n \left(1 + \frac{k}{n}\right)^3 \cdot \frac{1}{n} = \frac{1}{n^4} \sum_{k=1}^n (n+k)^3 \\
 &= \frac{1}{n^4} \frac{n^2}{4} (3 + 14n + 15n^2) \\
 &= \frac{1}{4} \left(\frac{3}{n^2} + \frac{14}{n} + 15 \right) //
 \end{aligned}$$

$$S = \lim_{n \rightarrow \infty} S_n = \frac{15}{4} //$$

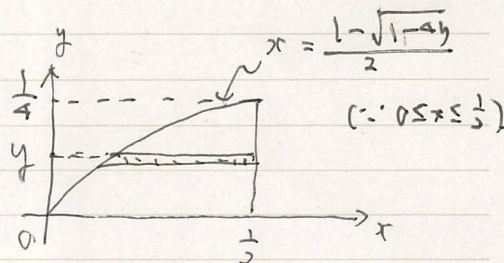
Prob #8

$$\begin{aligned}
 (a) \quad \Delta V(x) &= 2\pi x f(x) \Delta x \\
 &= 2\pi x (x - x^2) \Delta x.
 \end{aligned}$$



$$V = \lim \sum \Delta V(x) = \int_0^{1/2} 2\pi x (x - x^2) dx = \frac{5\pi}{96} //$$

$$\begin{aligned}
 (b) \quad \Delta V(y) &= \left[\pi \left(\frac{1}{2}\right)^2 - \pi \left(\frac{1 - \sqrt{1-4y}}{2}\right)^2 \right] \Delta y \\
 &= \frac{\pi}{4} (4y - 1 + 2\sqrt{1-4y}) \Delta y
 \end{aligned}$$



$$\begin{aligned}
 V &= \int_{(0,1)} dV = \int_0^{1/4} \frac{\pi}{4} (4y - 1 + 2\sqrt{1-4y}) dy \\
 &= \frac{5\pi}{96} //
 \end{aligned}$$